Group23_BrysonD_GorM

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Loading Data In

```
BBstats <- read.csv("statcast_data.csv")
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#Currently Data is WAYYY to large to parse through so im going to filter for year 2015 and onward

BBstats <- BBstats %>% filter(game_year >= 2015)
head(BBstats)
```

```
##    game_date game_year       player_name batter pitcher    events   description
## 1 2015-03-04      2015        Rios, Alex 425567  451596 field_out hit_into_play
## 2 2015-03-04      2015 Cuthbert, Cheslor 596144  592346   sac_fly hit_into_play
## 3 2015-03-04      2015     Hundley, Nick 460026  502624 field_out hit_into_play
## 4 2015-03-04      2015  Morales, Kendrys 434778  451596 field_out hit_into_play
## 5 2015-03-04      2015    Sierra, Moisés 501213  592346    double hit_into_play
## 6 2015-03-04      2015  Colón, Christian 518568  407890 field_out hit_into_play
##   game_type home_team away_team      pitch_name bat_speed release_speed
## 1         S       TEX        KC          Sinker        NA          90.1
## 2         S       TEX        KC       Curveball        NA          79.7
## 3         S       COL        AZ            <NA>        NA            NA
## 4         S       TEX        KC          Sinker        NA          89.4
## 5         S       TEX        KC 4-Seam Fastball        NA          91.4
## 6         S       TEX        KC 4-Seam Fastball        NA          87.5
##   release_spin_rate api_break_z_with_gravity api_break_x_arm
## 1                NA                     1.76           0.81
## 2                NA                     4.00          -0.59
## 3                NA                       NA             NA
## 4                NA                     1.82           1.00
## 5                NA                     1.77           0.17
```

```
## 6                NA                     1.33           0.73
##   api_break_x_batter_in launch_speed launch_angle hit_distance_sc
## 1                  0.81           NA           NA              NA
## 2                 -0.59           NA           NA              NA
## 3                    NA           NA           NA              NA
## 4                 -1.00           NA           NA              NA
## 5                  0.17           NA           NA              NA
## 6                  0.73           NA           NA              NA
##   estimated_ba_using_speedangle delta_run_exp
## 1                            NA        -0.221
## 2                            NA        -0.051
## 3                            NA        -0.230
## 4                            NA        -0.215
## 5                            NA         0.778
## 6                            NA        -0.255
```

```
str(BBstats)
```

```
## 'data.frame':    2316497 obs. of  22 variables:
##  $ game_date                   : chr  "2015-03-04" "2015-03-04" "2015-03-04" "2015-03-04" ...
##  $ game_year                   : int  2015 2015 2015 2015 2015 2015 2015 2015 2015 2015 ...
##  $ player_name                 : chr  "Rios, Alex" "Cuthbert, Cheslor" "Hundley, Nick" "Morales, Ker
##  $ batter                      : int  425567 596144 460026 434778 501213 518568 506747 543333 45612
##  $ pitcher                     : int  451596 592346 502624 451596 592346 407890 570810 451596 40789
##  $ events                      : chr  "field_out" "sac_fly" "field_out" "field_out" ...
##  $ description                 : chr  "hit_into_play" "hit_into_play" "hit_into_play" "hit_into_play
##  $ game_type                   : chr  "S" "S" "S" "S" ...
##  $ home_team                   : chr  "TEX" "TEX" "COL" "TEX" ...
##  $ away_team                   : chr  "KC" "KC" "AZ" "KC" ...
##  $ pitch_name                  : chr  "Sinker" "Curveball" NA "Sinker" ...
##  $ bat_speed                   : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ release_speed               : num  90.1 79.7 NA 89.4 91.4 87.5 83.9 91.1 88.1 92.1 ...
##  $ release_spin_rate           : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ api_break_z_with_gravity    : num  1.76 4 NA 1.82 1.77 1.33 2.76 1.48 1.43 2.03 ...
##  $ api_break_x_arm             : num  0.81 -0.59 NA 1 0.17 0.73 0.8 1.16 0.51 0.23 ...
##  $ api_break_x_batter_in       : num  0.81 -0.59 NA -1 0.17 0.73 0.8 -1.16 0.51 0.23 ...
##  $ launch_speed                : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ launch_angle                : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ hit_distance_sc             : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ estimated_ba_using_speedangle: num  NA NA NA NA NA NA NA NA NA NA ...
##  $ delta_run_exp               : num  -0.221 -0.051 -0.23 -0.215 0.778 -0.255 -0.269 0.731 -0.402 -
```

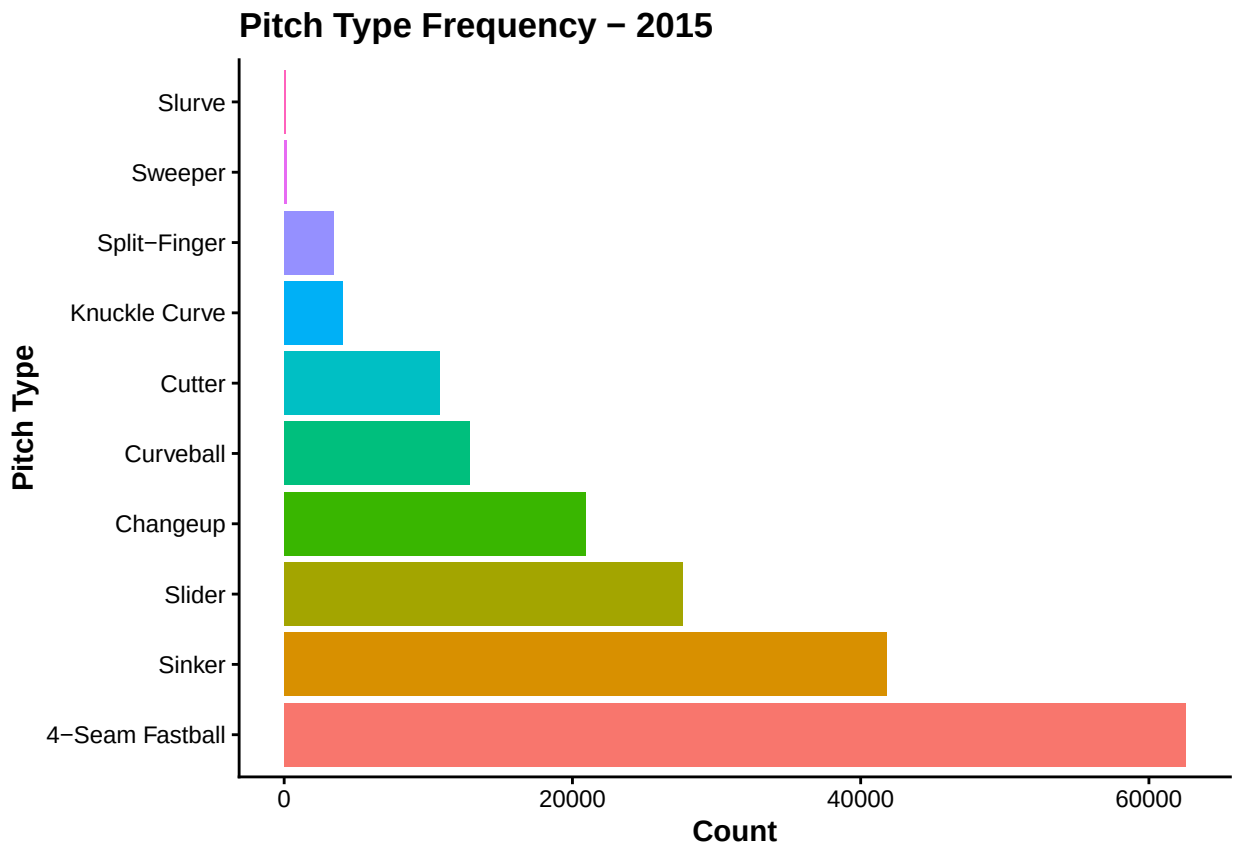
Frequency Chart of pitch types for the last 10 years. (most frequent pitches in 2015 all the way up to 2025)

```
BBstats1 <- BBstats %>% filter(game_type %in% c("R", "D", "L", "W")) %>%
  filter(pitch_name %in% c("Slurve" , "Knuckle Curve", "Split-Finger", "Curveball" , "Cutter" , "Sweepe
  filter(game_year %in% c(2015, 2018, 2021, 2023, 2025)) %>%
  group_by(game_year, pitch_name) %>%
  summarise(count = n(), .groups = "drop")

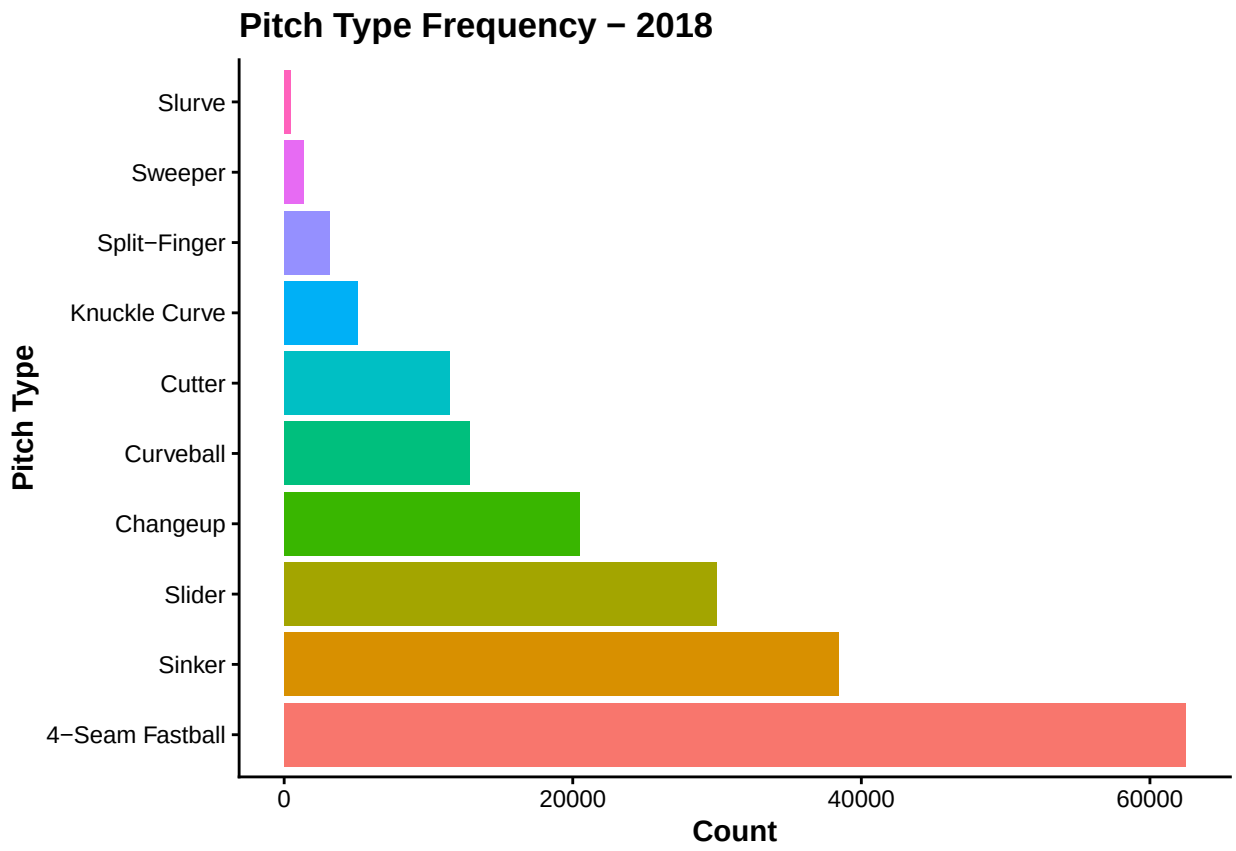
# Using a function to build a plot for every year instead of doing facet wrap because its easier to vie
pitch_freq_plot <- function(data, year) {
  data %>%
```

```
    filter(game_year == year) %>%
    mutate(pitch_name = reorder(pitch_name, -count)) %>%
    ggplot(aes(x = pitch_name, y = count, fill = pitch_name)) +
    geom_bar(stat = "identity") +
    coord_flip() +
    theme_classic() +
    theme(
      axis.text.y = element_text(size = 9),  # y instead of x now
      axis.text.x = element_text(size = 9),
      legend.position = "none",
      axis.title = element_text(face = "bold"),
      plot.title = element_text(face = "bold", size = 13)
    ) +
    labs(
      title = paste("Pitch Type Frequency -", year),
      x = "Pitch Type",
      y = "Count"
    )
}

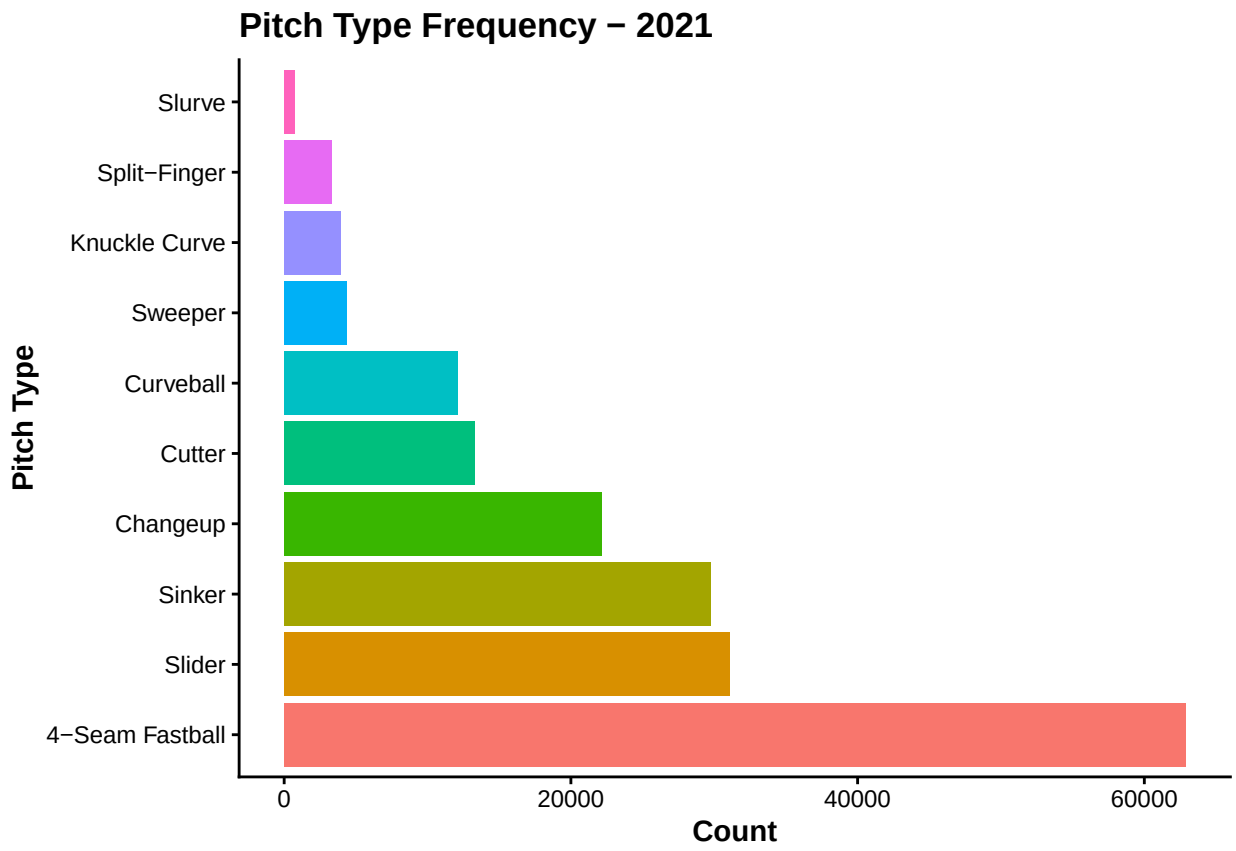
plot_2015 <- pitch_freq_plot(BBstats1, 2015)
plot_2018 <- pitch_freq_plot(BBstats1, 2018)
plot_2021 <- pitch_freq_plot(BBstats1, 2021)
plot_2023 <- pitch_freq_plot(BBstats1, 2023)
plot_2025 <- pitch_freq_plot(BBstats1, 2025)
plot_2015
```


Pitch Type Frequency − 2015
Slurve
Sweeper
Split−Finger
Knuckle Curve
Cutter
Curveball
Changeup
Slider
Sinker
4−Seam Fastball
Pitch Type
0
20000
40000
60000
Count

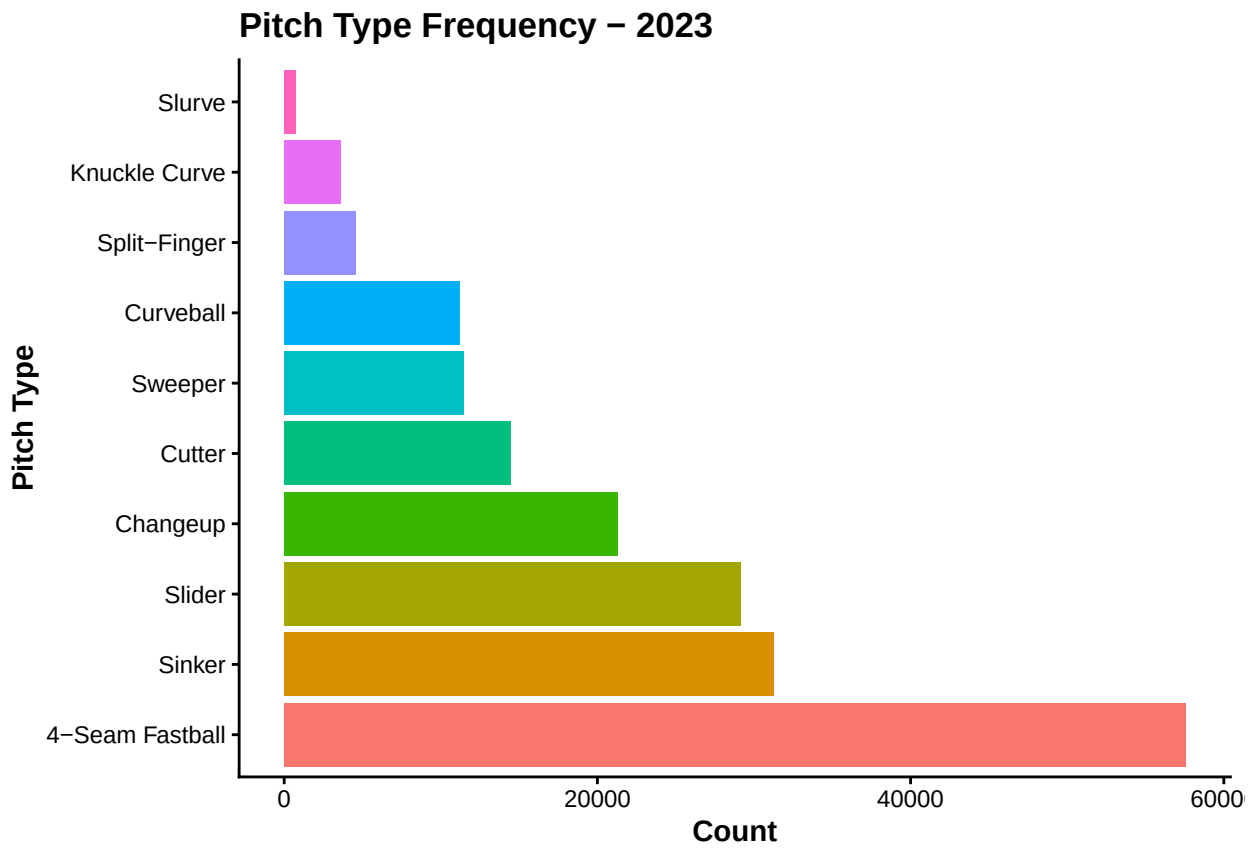
```
plot_2018
```


Pitch Type Frequency − 2018
Slurve
Sweeper
Split−Finger
Knuckle Curve
Cutter
Curveball
Changeup
Slider
Sinker
4−Seam Fastball
Pitch Type
0
20000
40000
60000
Count

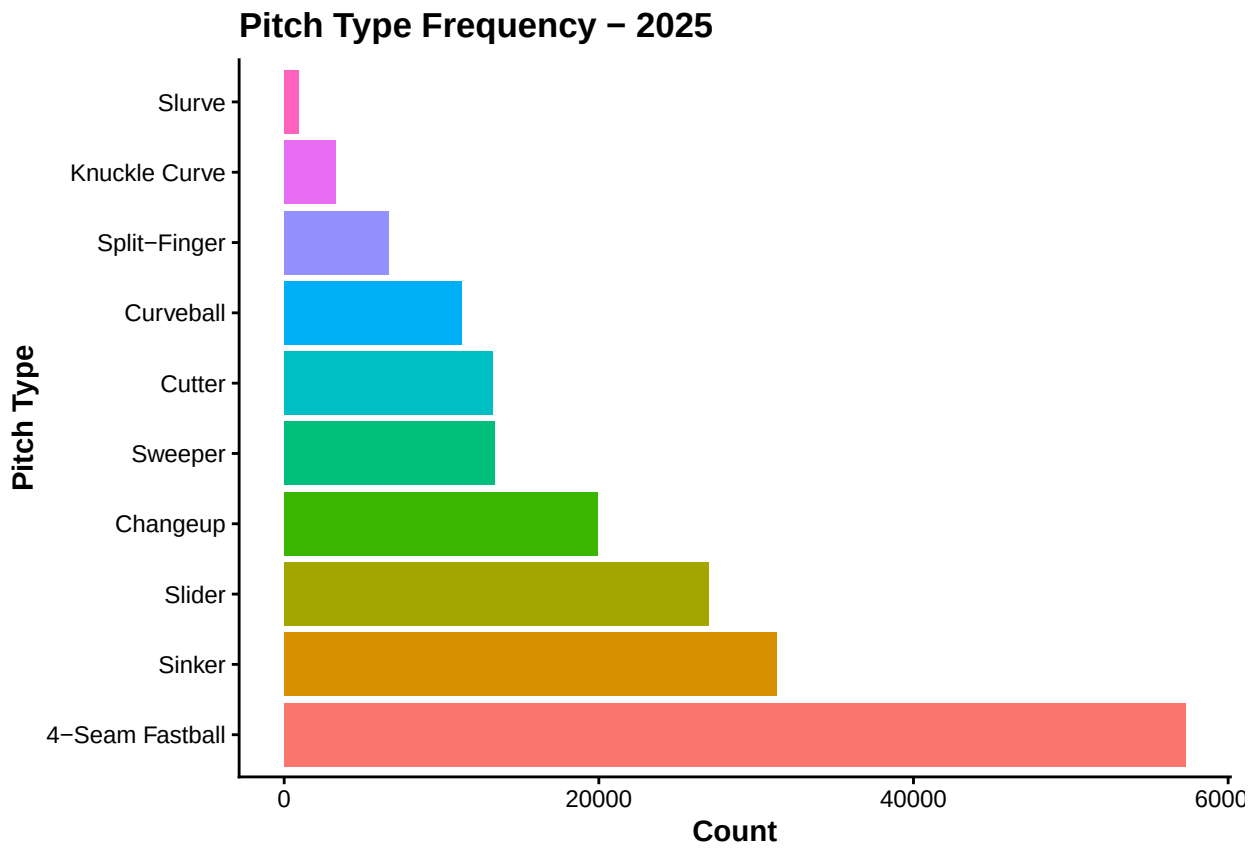
```
plot_2021
```


Pitch Type Frequency − 2021
Slurve
Split−Finger
Knuckle Curve
Sweeper
Curveball
Cutter
Changeup
Sinker
Slider
4−Seam Fastball
Pitch Type
0
20000
40000
60000
Count

```
plot_2023
```


Pitch Type Frequency − 2023
Slurve
Knuckle Curve
Split−Finger
Curveball
Sweeper
Cutter
Changeup
Slider
Sinker
4−Seam Fastball
Pitch Type
0
20000
40000
6000
Count

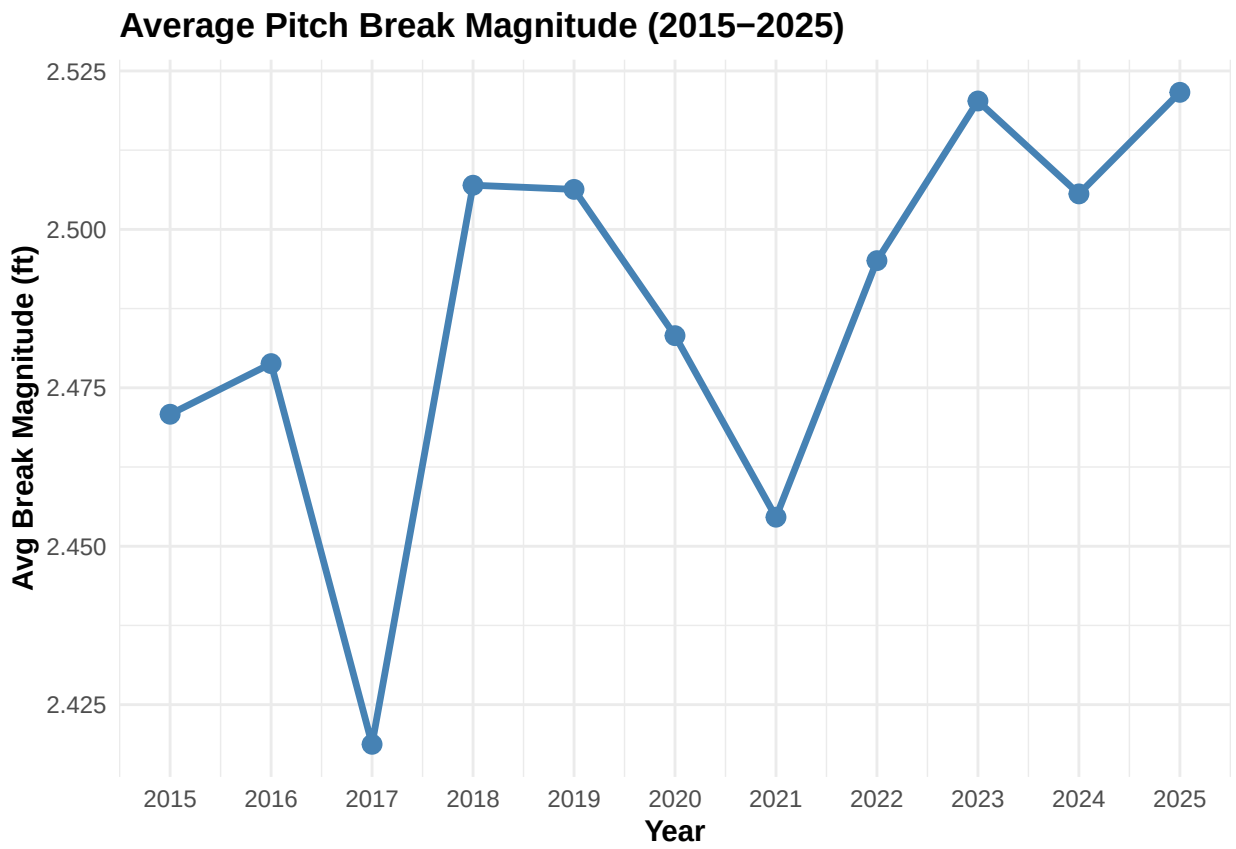
```
plot_2025
```



Build a Breaking ball vector then calculate the magnitude and observe its changes over the last 10 years

```
BBstats_break <- BBstats %>%
  filter(game_type %in% c("R", "D", "L", "W")) %>%
  filter(!is.na(api_break_x_arm), !is.na(api_break_z_with_gravity)) %>%
  mutate(break_magnitude = sqrt(api_break_x_arm^2 + api_break_z_with_gravity^2)) %>%
  group_by(game_year) %>%
  summarise(avg_break = mean(break_magnitude), .groups = "drop")

ggplot(BBstats_break, aes(x = game_year, y = avg_break)) +
  geom_line(linewidth = 1.2, color = "steelblue") +
  geom_point(size = 3, color = "steelblue") +
  theme_minimal() +
  theme(
    axis.title = element_text(face = "bold"),
    plot.title = element_text(face = "bold", size = 13)
  ) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 1)) +
  labs(
    title = "Average Pitch Break Magnitude (2015-2025)",
    x = "Year",
    y = "Avg Break Magnitude (ft)"
  )
```




```
cor.test(BBstats_break$game_year, BBstats_break$avg_break)
```

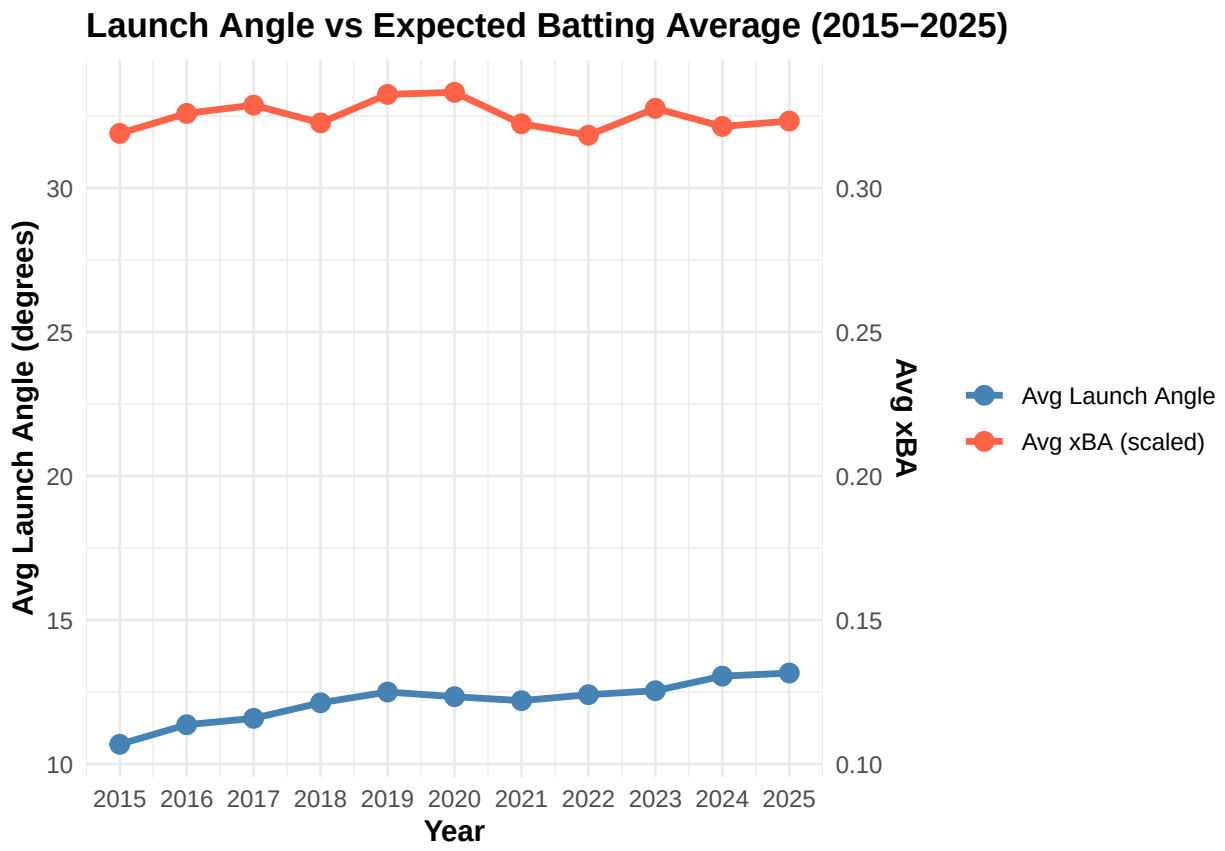
```
##
##  Pearson's product-moment correlation
##
## data:  BBstats_break$game_year and BBstats_break$avg_break
## t = 2.1147, df = 9, p-value = 0.06358
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.03626334  0.87396465
## sample estimates:
##       cor
## 0.5761448
```

```
#p-value = 0.06358
# 95 percent confidence interval: -0.03626334 - 0.87396465
# cor = 0.5761448
```

Batting Average vs Launch Angle ? (Run Statistical Test, What type of correlation)? The higher people hit the ball vs how sucessful they are what we expect –> people hit the ball higher and will have a lower batting average, this is because in recent years more hitters have beeing going for a home run or bust mentality, this means that they prob hit more home runs but also have a lower batting average. How can I calcualte this change over the years.

```r
BBstats_angle <- BBstats %>%
  filter(game_type %in% c("R", "D", "L", "W")) %>%
  filter(!is.na(launch_angle), !is.na(estimated_ba_using_speedangle)) %>%
  group_by(game_year) %>%
  summarise(
    avg_launch_angle = mean(launch_angle),
    avg_xba = mean(estimated_ba_using_speedangle),
    .groups = "drop"
  )

ggplot(BBstats_angle, aes(x = game_year)) +
  geom_line(aes(y = avg_launch_angle, color = "Avg Launch Angle"), linewidth = 1.2) +
  geom_point(aes(y = avg_launch_angle, color = "Avg Launch Angle"), size = 3) +
  geom_line(aes(y = avg_xba * 100, color = "Avg xBA (scaled)"), linewidth = 1.2) +
  geom_point(aes(y = avg_xba * 100, color = "Avg xBA (scaled)"), size = 3) +
  scale_y_continuous(
    name = "Avg Launch Angle (degrees)",
    sec.axis = sec_axis(~ . / 100, name = "Avg xBA")
  ) +
  scale_x_continuous(breaks = seq(2015, 2025, by = 1)) +
  scale_color_manual(values = c("Avg Launch Angle" = "steelblue", "Avg xBA (scaled)" = "tomato")) +
  theme_minimal() +
  theme(
    axis.title = element_text(face = "bold"),
    plot.title = element_text(face = "bold", size = 13),
    legend.title = element_blank()
  ) +
  labs(title = "Launch Angle vs Expected Batting Average (2015-2025)",
       x = "Year")
```




```
BBstats_raw <- BBstats %>%
  filter(game_type %in% c("R", "D", "L", "W")) %>%
  filter(!is.na(launch_angle), !is.na(estimated_ba_using_speedangle))
cor.test(BBstats_raw$launch_angle, BBstats_raw$estimated_ba_using_speedangle)
```

```
##
##  Pearson's product-moment correlation
##
## data:  BBstats_raw$launch_angle and BBstats_raw$estimated_ba_using_speedangle
## t = -26.803, df = 1287804, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.02533834 -0.02188603
## sample estimates:
##         cor
## -0.02361225
```

r = -0.0236 — negative direction, confirming your hypothesis directionally p < 2.2e-16 — essentially zero, statistically bulletproof with 1.28 million observations 95% CI: (-0.0253, -0.0219) — extremely tight, meaning this negative relationship is real and not noise

```
library(tidyverse)

baseball_data <- read.csv("statcast_data.csv")
head(baseball_data)
```

```
##    game_date game_year       player_name batter pitcher    events   description
## 1 2015-03-04      2015        Rios, Alex 425567  451596 field_out hit_into_play
## 2 2015-03-04      2015 Cuthbert, Cheslor 596144  592346   sac_fly hit_into_play
## 3 2015-03-04      2015     Hundley, Nick 460026  502624 field_out hit_into_play
## 4 2015-03-04      2015  Morales, Kendrys 434778  451596 field_out hit_into_play
## 5 2015-03-04      2015    Sierra, Moisés 501213  592346    double hit_into_play
## 6 2015-03-04      2015  Colón, Christian 518568  407890 field_out hit_into_play
##   game_type home_team away_team      pitch_name bat_speed release_speed
## 1         S       TEX        KC          Sinker        NA          90.1
## 2         S       TEX        KC       Curveball        NA          79.7
## 3         S       COL        AZ            <NA>        NA            NA
## 4         S       TEX        KC          Sinker        NA          89.4
## 5         S       TEX        KC 4-Seam Fastball        NA          91.4
## 6         S       TEX        KC 4-Seam Fastball        NA          87.5
##   release_spin_rate api_break_z_with_gravity api_break_x_arm
## 1                NA                     1.76            0.81
## 2                NA                     4.00           -0.59
## 3                NA                       NA              NA
## 4                NA                     1.82            1.00
## 5                NA                     1.77            0.17
## 6                NA                     1.33            0.73
##   api_break_x_batter_in launch_speed launch_angle hit_distance_sc
## 1                  0.81           NA           NA              NA
## 2                 -0.59           NA           NA              NA
## 3                    NA           NA           NA              NA
## 4                 -1.00           NA           NA              NA
## 5                  0.17           NA           NA              NA
## 6                  0.73           NA           NA              NA
##   estimated_ba_using_speedangle delta_run_exp
## 1                            NA        -0.221
## 2                            NA        -0.051
## 3                            NA        -0.230
## 4                            NA        -0.215
## 5                            NA         0.778
## 6                            NA        -0.255
```

```
# Extract the types of events that have occurred and been recorded over the past 10 years
list_of_events <- unique(baseball_data$events)
list_of_events
```

```
##  [1] "field_out"                "sac_fly"
##  [3] "double"                   "fielders_choice_out"
##  [5] "triple"                   "force_out"
##  [7] "walk"                     "field_error"
##  [9] "strikeout"                "single"
## [11] "home_run"                 "hit_by_pitch"
## [13] "grounded_into_double_play" "fielders_choice"
## [15] "truncated_pa"             "double_play"
## [17] "sac_bunt"                 "sac_fly_double_play"
## [19] "strikeout_double_play"    "intent_walk"
## [21] "sac_bunt_double_play"     "triple_play"
## [23] "catcher_interf"           "ejection"
## [25] "game_advisory"
```

```r
# Create a vector of hit events
hit_events <- c("single", "double", "triple", "home_run")

# Create a vector of non-AB events (don't count towards batting average)
non_ab_events <- c("walk", "intent_walk", "hit_by_pitch", "sac_fly", "sac_bunt",
                   "sac_bunt_double_play", "sac_fly_double_play", "truncated_pa",
                   "catcher_interf", "game_advisory", "ejection")

# Subsetting data to only include regular season and playoff games
game_types <- unique(baseball_data$game_type)
game_types
```

```
## [1] "S"     "E"     "R"     "A"     "F"     "D"     "L"     "W"     "FALSE"
```

```r
# Vector of only regular season and playoff game types
reg_playoff <- c("R", "D", "L", "W")

# Subsetting data to only include regular season and playoff games
baseball_data <- baseball_data |>
  filter(game_type %in% reg_playoff)
```

```r
# Creating a dataset with only PAs that count as ABs
AB_only_data <- baseball_data |>
  # Filtering out events that do not count as an AB
  filter(!(events %in% non_ab_events))

# Creating a summary of aggregated batting average per year
BA_overall_summary <- AB_only_data |>
  group_by(game_year) |>
  # Calculate each year's average batting average
  summarize(batting_average = mean(events %in% hit_events) |> round(digits = 4))

BA_overall_summary
```

```
## # A tibble: 11 x 2
##    game_year batting_average
##        <int>           <dbl>
##  1      2015           0.254
##  2      2016           0.255
##  3      2017           0.254
##  4      2018           0.248
##  5      2019           0.252
##  6      2020           0.244
##  7      2021           0.244
##  8      2022           0.242
##  9      2023           0.248
## 10      2024           0.243
## 11      2025           0.245
```

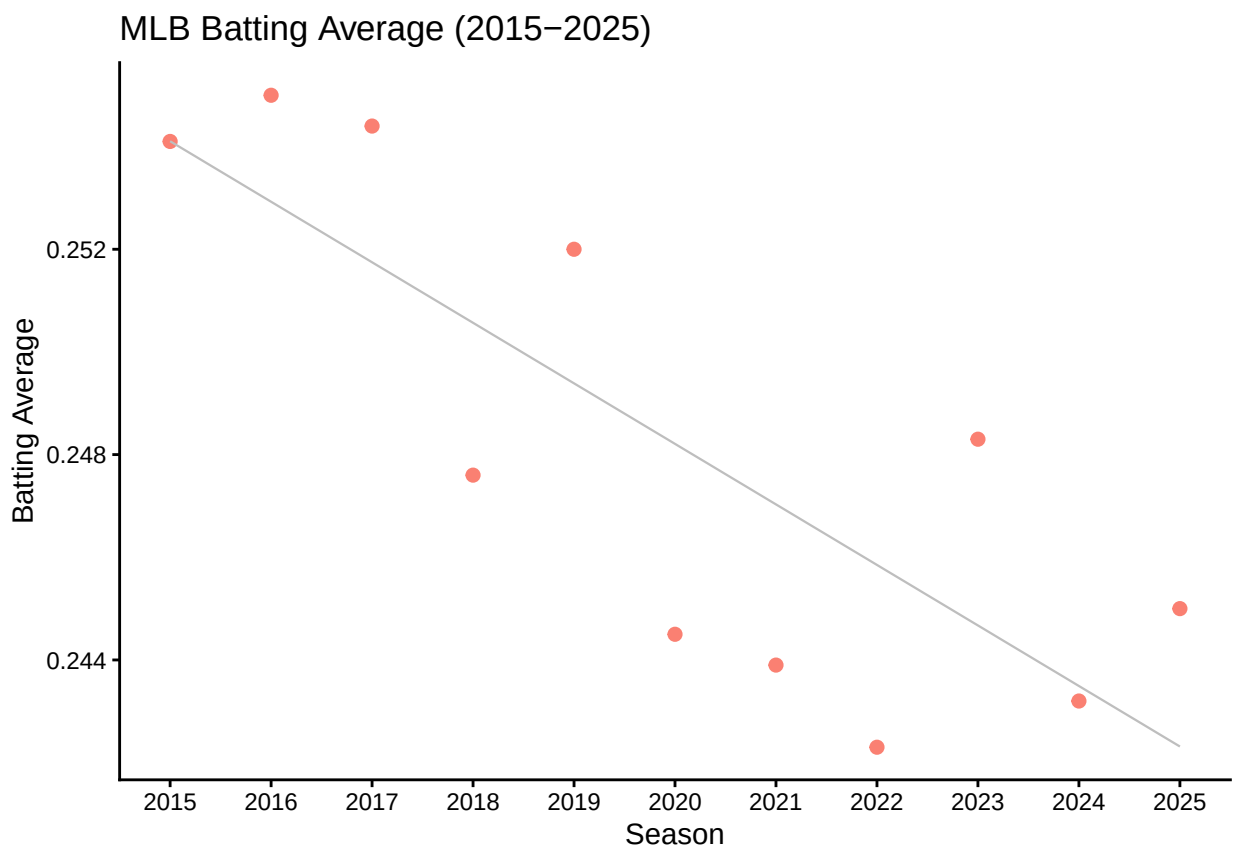
```r
# Creating a scatter plot of overall batting average versus year
BA_overall_plot <- ggplot(BA_overall_summary, aes(x = game_year, y = batting_average)) +
  # Plotting the points
```

```
  geom_point(color = "salmon", size = 2) +
  # Plotting a line of best fit
  geom_smooth(size = 0.4, method = "lm", se = FALSE, color = "grey") +
  # Forcing the x-axis into one-year breaks
  scale_x_continuous(breaks = 2015:2025) +
  labs(title = "MLB Batting Average (2015-2025)",
       x = "Season",
       y = "Batting Average") +
  theme_classic()
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
BA_overall_plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```




```
# Computing linear regression statistics on batting average versus year
BA_linear_regression <- lm(batting_average ~ game_year, data = BA_overall_summary)
summary(BA_linear_regression)
```

```
##
## Call:
## lm(formula = batting_average ~ game_year, data = BA_overall_summary)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0037091 -0.0030486 -0.0000045  0.0026327  0.0036282
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.6299727  0.5870026   4.480  0.00153 **
## game_year   -0.0011791  0.0002906  -4.058  0.00285 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003048 on 9 degrees of freedom
## Multiple R-squared:  0.6466, Adjusted R-squared:  0.6073
## F-statistic: 16.46 on 1 and 9 DF,  p-value: 0.002853
```

```
# Analyzing all types of pitches in the AB only data
unique(AB_only_data$pitch_name)
```

```
##  [1] "Cutter"          "4-Seam Fastball" "Curveball"       "Slider"
##  [5] "Changeup"        "Sinker"          "Knuckle Curve"   "Eephus"
##  [9] NA                "Split-Finger"    "Knuckleball"     "Sweeper"
## [13] "Forkball"        "Slurve"          "Other"           "Slow Curve"
## [17] "Screwball"       "Automatic Ball"  "Unknown"
```

```
# Deducing the total number of pitches/proportion of each type from 2015-2025
pitch_count <- AB_only_data |>
  group_by(pitch_name) |>
  summarize(total = n(), prop = round(n()/nrow(AB_only_data), 3)) |>
  # Arranging in order from most common to least common pitch types
  arrange(desc(total))

print(pitch_count)
```

```
## # A tibble: 19 x 3
##    pitch_name       total  prop
##    <chr>            <int> <dbl>
##  1 4-Seam Fastball 548184 0.317
##  2 Sinker          317217 0.183
##  3 Slider          278769 0.161
##  4 Changeup        200994 0.116
##  5 Curveball       120849 0.07
##  6 Cutter          117917 0.068
##  7 Sweeper          51292 0.03
##  8 Knuckle Curve    39901 0.023
##  9 Split-Finger     38914 0.022
## 10 Slurve            5654 0.003
## 11 <NA>              3353 0.002
## 12 Knuckleball       3109 0.002
## 13 Other             1549 0.001
```

```
## 14 Eephus            1168 0.001
## 15 Forkball           660 0
## 16 Slow Curve         270 0
## 17 Screwball           85 0
## 18 Automatic Ball       1 0
## 19 Unknown              1 0
```

```
# Deducing the total percentage of pitches thrown that the
# six most common pitches make up
common_pitches_prop <- sum(pitch_count$prop[1:6])
common_pitches_prop
```

```
## [1] 0.915
```

```
# Creating a vector of the six most common pitch types
desired_pitches <- c()
for(i in 1:6){
  desired_pitches[i] <- pitch_count$pitch_name[i]
}
```

```
# Creating a summary of batting average per pitch per year
BA_by_pitch_summary <- AB_only_data |>
  # Choosing the desired (six most common) pitches
  filter(pitch_name %in% desired_pitches) |>
  group_by(game_year, pitch_name) |>
  # Calculating/summarizing the batting average per pitch type
  summarize(batting_average = mean(events %in% hit_events) |> round(digits = 4))
```

```
## `summarise()` has grouped output by 'game_year'. You can override using the
## `.groups` argument.
```

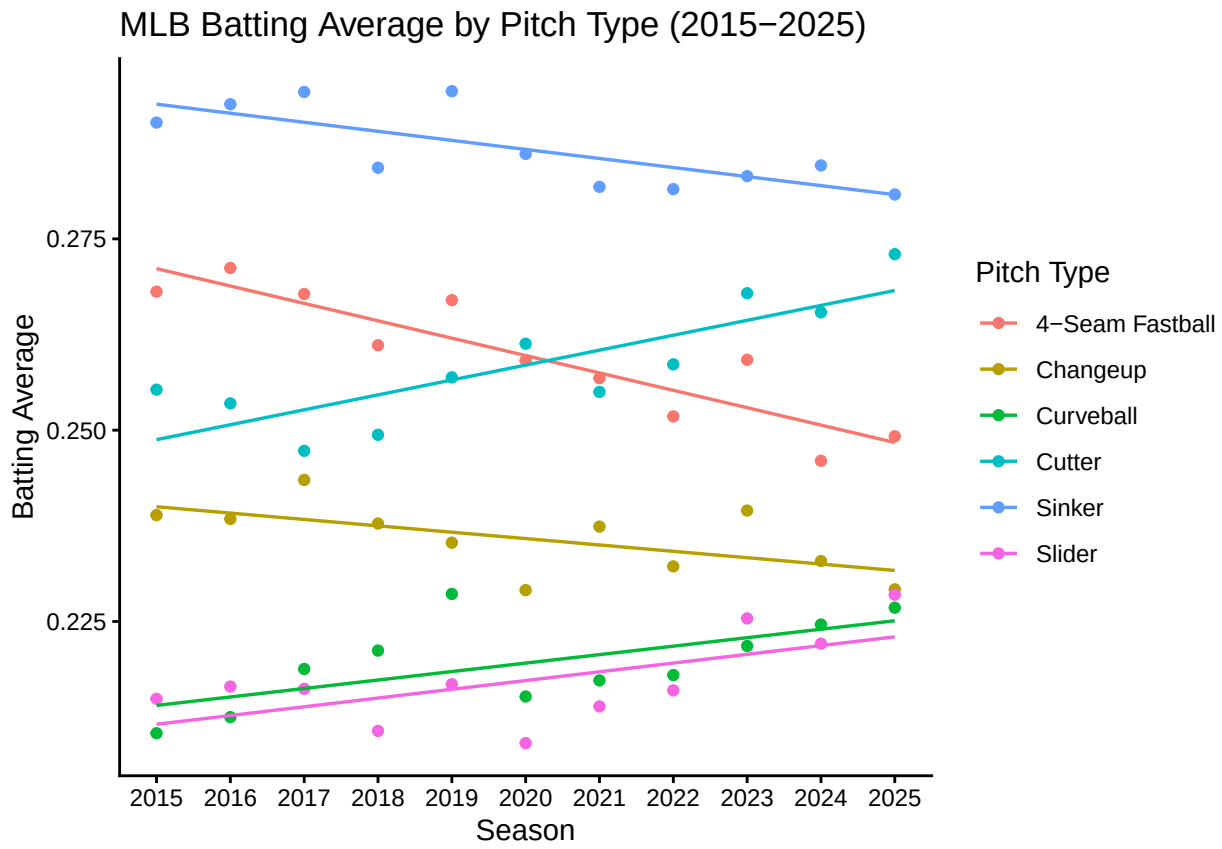
```
BA_by_pitch_summary
```

```
## # A tibble: 66 x 3
## # Groups:   game_year [11]
##    game_year pitch_name      batting_average
##        <int> <chr>                     <dbl>
##  1      2015 4-Seam Fastball           0.268
##  2      2015 Changeup                  0.239
##  3      2015 Curveball                 0.210
##  4      2015 Cutter                    0.255
##  5      2015 Sinker                    0.290
##  6      2015 Slider                    0.215
##  7      2016 4-Seam Fastball           0.271
##  8      2016 Changeup                  0.238
##  9      2016 Curveball                 0.212
## 10      2016 Cutter                    0.254
## # i 56 more rows
```

```
# Plotting the batting average per pitch type
ggplot(BA_by_pitch_summary, aes(x = game_year, y = batting_average, color = pitch_name)) +
```

```
  geom_point(size = 1.5) +
  scale_x_continuous(breaks = 2015:2025) +
  geom_smooth(size = 0.6, method = "lm", se = FALSE) +
  labs(title = "MLB Batting Average by Pitch Type (2015-2025)",
       x = "Season",
       y = "Batting Average",
       color = "Pitch Type") +
  theme_classic()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Filtering out data that does not have estimated/expected (xBA) batting average values

baseball_data <- rename(baseball_data, xBA = estimated_ba_using_speedangle)
 xBA_baseball_data <- baseball_data |>
   filter(!is.na(xBA))

# Creating a new column that displays TRUE if the event was a hit, FALSE otherwise
xBA_baseball_data$is_hit <- xBA_baseball_data$events %in% hit_events

# Summarizing the aggregated xBA versus actual BA for 2015-2025
xBA_BA_summary <- xBA_baseball_data |>
  group_by(game_year) |>
  summarize(expected_batting_average = round(mean(xBA), 4),
```

```
            actual_batting_average = round(mean(is_hit), 4),
            # Adding an actual BA minus xBA statistic for each year
            actual_minus_expected = actual_batting_average - expected_batting_average) |>
  # Combining xBA and actual BA into one column for data visualization
  pivot_longer(
    cols = c(expected_batting_average, actual_batting_average),
    names_to = "BA_Type",
    values_to = "BA"
  )

xBA_BA_summary
```

```
## # A tibble: 22 x 4
##    game_year actual_minus_expected BA_Type                    BA
##        <int>                 <dbl> <chr>                   <dbl>
##  1      2015                0.0086  expected_batting_average 0.319
##  2      2015                0.0086  actual_batting_average   0.328
##  3      2016                0.00700 expected_batting_average 0.326
##  4      2016                0.00700 actual_batting_average   0.333
##  5      2017                0.00680 expected_batting_average 0.329
##  6      2017                0.00680 actual_batting_average   0.336
##  7      2018                0.00680 expected_batting_average 0.323
##  8      2018                0.00680 actual_batting_average   0.329
##  9      2019                0.00630 expected_batting_average 0.332
## 10      2019                0.00630 actual_batting_average   0.339
## # i 12 more rows
```

```
# Five number summary for the actual_minus_expected BA values
delta_summary <- summary(xBA_BA_summary$actual_minus_expected)
delta_summary
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.000600 0.004325 0.006300 0.005655 0.006950 0.008600
```

```
# Checking for statistical outlier actual_minus_expected data
Q1 <- delta_summary[2]; Q3 <- delta_summary[5]
outlier_lower_bound <- Q1 - 1.5 * (Q3 - Q1)
outlier_lower_bound
```

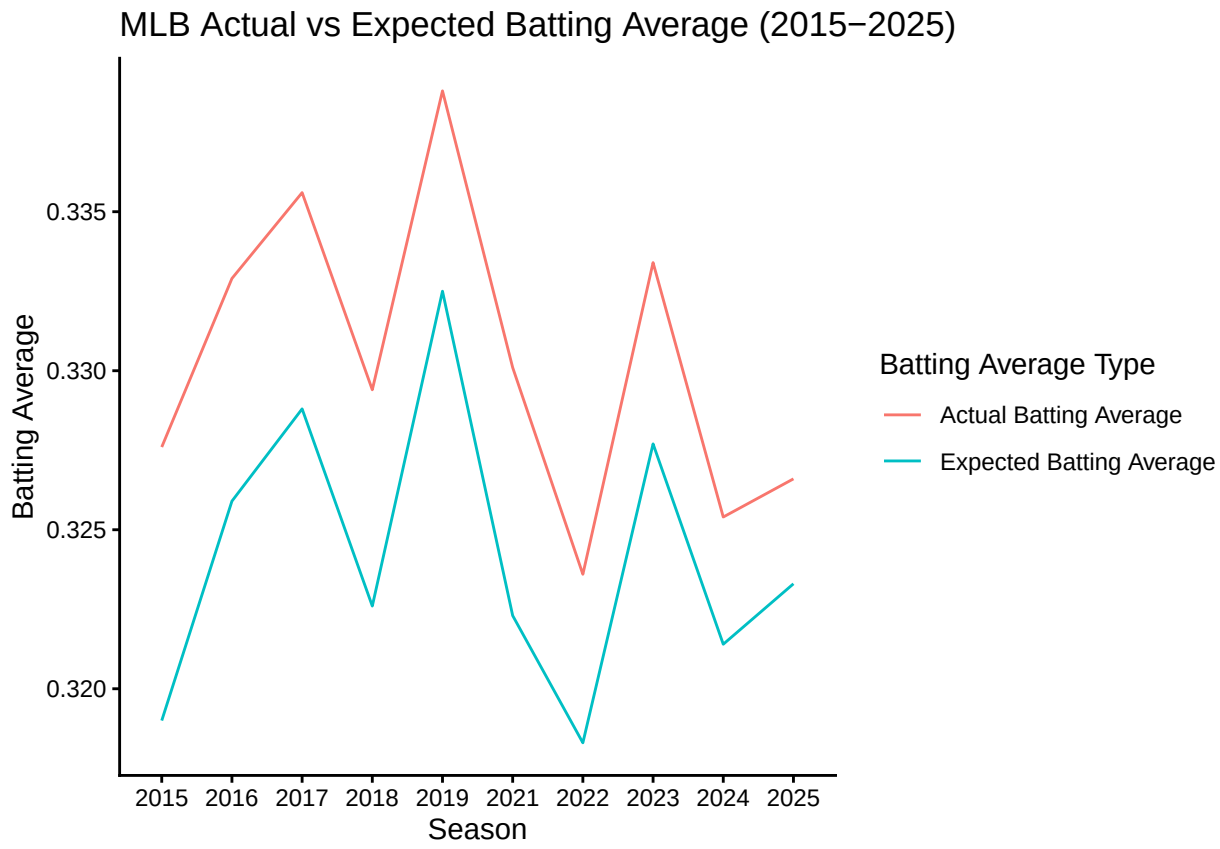
```
##   1st Qu.
## 0.0003875
```

```
# Filtering out the 2020 outlying actual_minus_expected statistic
xBA_BA_summary <- xBA_BA_summary |>
  filter(game_year != 2020)

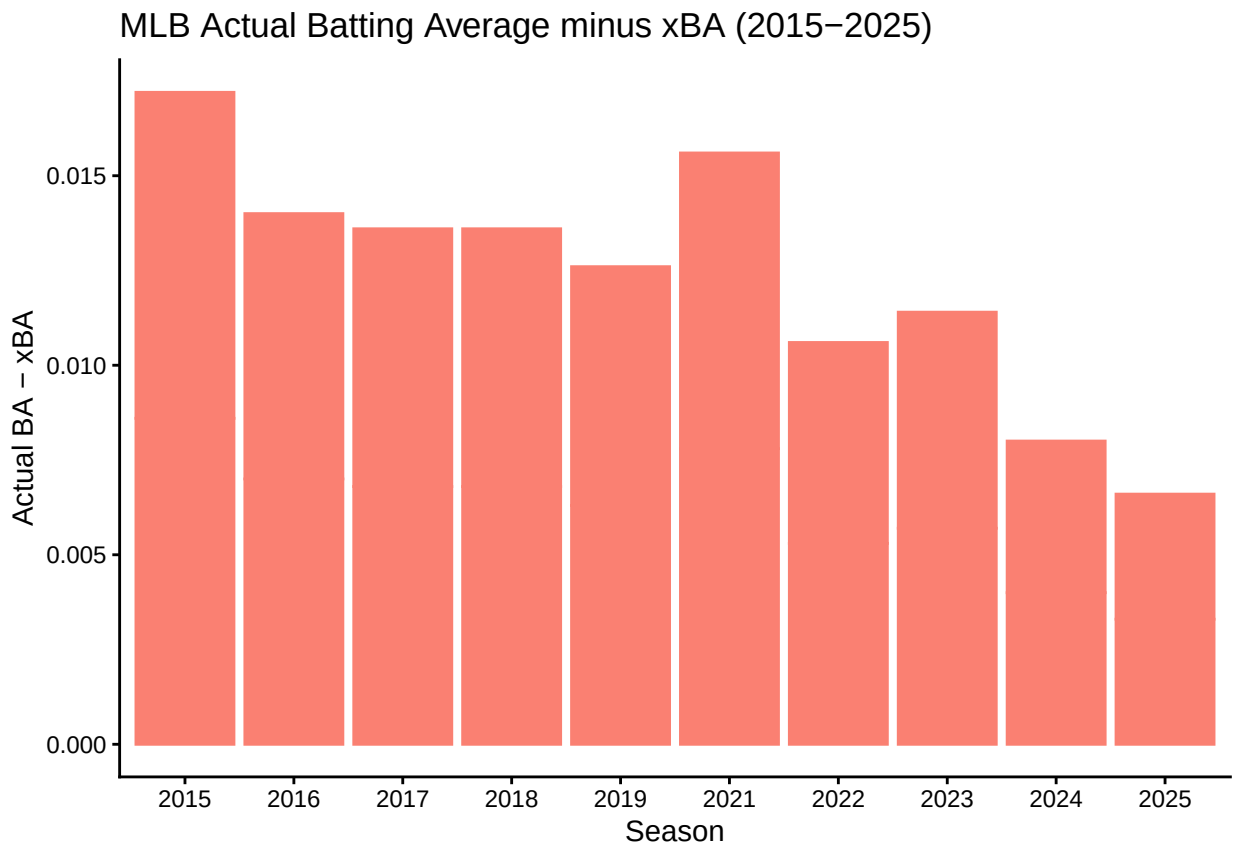
# Plotting the two types of batting averages against the year
xBA_BA_plot <- ggplot(xBA_BA_summary, aes(x = as.factor(game_year), y = BA, color = BA_Type, group = BA
  geom_line() +
  labs(title = "MLB Actual vs Expected Batting Average (2015-2025)",
       x = "Season",
```

```
       y = "Batting Average",
       color = "Batting Average Type") +
  # Renaming the color labels
  scale_color_discrete(labels = c("Actual Batting Average", "Expected Batting Average")) +
  theme_classic()

xBA_BA_plot
```




```
# Plotting the actual_minus_expected BA values for each year, less 2020
delta_BA_plot <- ggplot(xBA_BA_summary, aes(x = as.factor(game_year), y = actual_minus_expected)) +
  # Creating a bar chart
  geom_col(fill = "salmon", color = "salmon") +
  labs(title = "MLB Actual Batting Average minus xBA (2015-2025)",
       x = "Season",
       y = "Actual BA - xBA",) +
  theme_classic()
delta_BA_plot
```



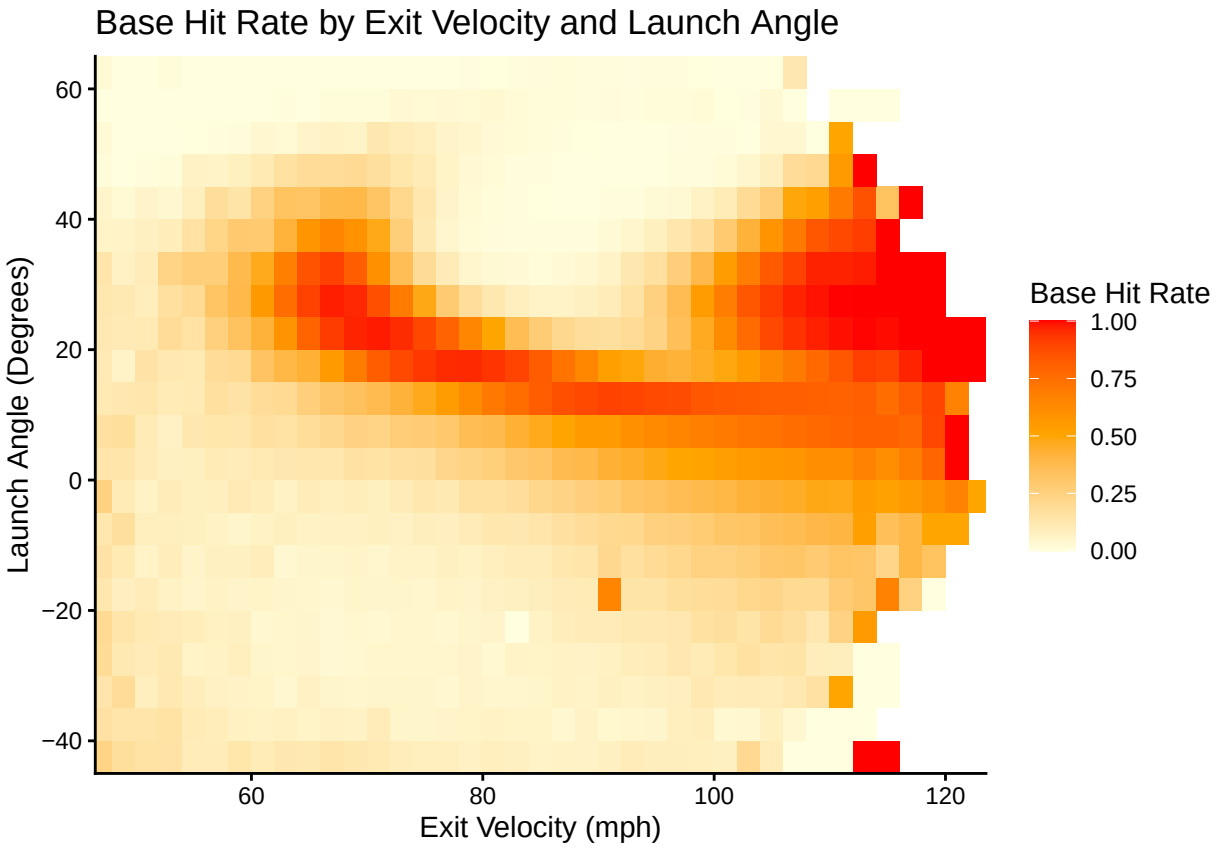

```
# Computing linear regression statistics on actual_minus_expected BA versus year
delta_BA_linear_regression <- lm(actual_minus_expected ~ game_year, data = xBA_BA_summary)
summary(delta_BA_linear_regression)
```

```
##
## Call:
## lm(formula = actual_minus_expected ~ game_year, data = xBA_BA_summary)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0008918 -0.0005854 -0.0002004  0.0004718  0.0020336
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.8013054  0.1195873   6.701 2.77e-06 ***
## game_year   -0.0003936  0.0000592  -6.649 3.07e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0008781 on 18 degrees of freedom
## Multiple R-squared:  0.7107, Adjusted R-squared:  0.6946
## F-statistic: 44.21 on 1 and 18 DF,  p-value: 3.067e-06
```

```
# Filtering out data that does not include launch speed and launch angle
batted_ball_data <- xBA_baseball_data |>
```

```
  filter(!is.na(launch_speed) & !is.na(launch_angle))

# Creating a heatmap for launch angle versus launch speed, colored based off proportion of base hits pe
batted_ball_heatmap <- ggplot(batted_ball_data, aes(x = launch_speed, y = launch_angle, z = as.numeric(
  # Sizing each unit of area to be 2mph of launch speed by 5 degrees of launch angle
  stat_summary_2d(fun = mean, binwidth = c(2, 5)) +
  scale_fill_gradientn(
    colors = c("lightyellow", "orange", "red"),
    name = "Base Hit Rate"
  ) +
  # Adjusting coordinates to exclude outlier values
  coord_cartesian(xlim = c(50, 120), ylim = c(-40, 60)) +
  labs(
    title = "Base Hit Rate by Exit Velocity and Launch Angle",
    x = "Exit Velocity (mph)",
    y = "Launch Angle (Degrees)"
  ) +
  theme_classic() +
  theme(
    panel.grid.minor = element_blank() # Removes minor gridlines for a cleaner look
  )
batted_ball_heatmap
```




```
# Computing generalized linear regression statistics on batting average versus year
batted_ball_model <- glm(as.numeric(is_hit) ~ launch_speed + launch_angle + I(launch_angle^2),
```

```
                data = batted_ball_data,
                family = "binomial")
summary(batted_ball_model)
```

```
##
## Call:
## glm(formula = as.numeric(is_hit) ~ launch_speed + launch_angle +
##     I(launch_angle^2), family = "binomial", data = batted_ball_data)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -2.681e+00  1.619e-02  -165.6   <2e-16 ***
## launch_speed       2.736e-02  1.725e-04   158.7   <2e-16 ***
## launch_angle       2.814e-02  1.532e-04   183.6   <2e-16 ***
## I(launch_angle^2) -1.277e-03  4.640e-06  -275.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1634268  on 1287805  degrees of freedom
## Residual deviance: 1403754  on 1287802  degrees of freedom
## AIC: 1403762
##
## Number of Fisher Scoring iterations: 6
```